

Mission Planner Customization Progress Report

Project Overview

Customized Mission Planner based Ground station application for Saif Seas with UI modernization, swarm mission support, telemetry dashboard enhancements, and operational workflow improvements.

Completed Tasks

1. UI/Branding Enhancements

Completed

- Updated application branding and visual theme.
- Modified software icons and resources to better match company color scheme.
- Added custom splash screen support.
- Added animated splash/loading behavior.
- Improved menu and panel organization.
- Added functionality to hide/show menu icons dynamically.
- Added eye-toggle style UI interactions.
- Reorganized HUD into separate tab structure.
- Added new telemetry-oriented Dynamics tab.

Result

Cleaner and more professional operator interface.

2. Telemetry Dashboard Implementation

Completed

- Created new telemetry dashboard tab (tabPage_Dynamics).
- Extracted telemetry data from existing HUD binding source.
- Implemented live telemetry bindings using bindingSourceHud.
- Added real-time telemetry display for:
 - Connection Status
 - Armed/Disarmed State
 - Flight Mode
 - Altitude
 - Ground Speed
 - Heading/Yaw

- Battery Status
- GPS Information
- Waypoint Information
- Added formatted telemetry display:
 - Connected/Disconnected
 - Armed/Disarmed
 - Colored status indicators

Result

Legacy HUD dependency reduced while improving readability and operator usability.

3. Vehicle Orientation Visualization

Completed

- Implemented heading/yaw-based vehicle image rotation.
- Added top-view vehicle visualization.
- Integrated live heading telemetry with graphical orientation display.
- Connected telemetry updates to real-time image rotation.

Result

Improved situational awareness and modernized visualization.

4. Swarm Mission / Multi-Plan Features

Completed

- Implemented waypoint duplication and offset generation.
- Added bearing-based mission shifting.
- Added distance-based mission generation.
- Implemented automatic generation of multiple shifted mission plans.
- Added mission save/export automation.
- Added automatic folder creation for grouped mission plans.
- Added dropdown-based mission loading system.
- Added directory scanning for .waypoints files.
- Added multi-plan visualization support.
- Added different colored markers/routes for different plans.
- Added marker selection/unselection logic.

- Added ctrl+A selection handling.
- Added deselection logic.
- Improved marker interaction behavior.
- Added ctrl+Z for undo logics with swarm functionality.

Result

Initial swarm mission planning workflow established.

5. Map Interaction Improvements

Completed

- Investigated and modified map mouse interaction handlers.
- Improved marker selection handling.
- Added group marker logic.
- Added keyboard interaction support.
- Improved waypoint interaction handling.
- Investigated route rendering pipeline.

Result

Better operator interaction and mission editing workflow.

6. Video Streaming Investigation

Completed

- Investigated GStreamer integration.
- Investigated VLC-based video streaming.
- Validated simulated video feed architecture.
- Evaluated UDP/HTTP streaming workflows.
- Studied future drone video integration pipeline.

Result

Video integration architecture understood and partially validated.

7. Project Backup / Version Control

Completed

- Configured workflow.
- Created repository backup.
- Configured commit workflow.

- Established version-control process.
- Added multiple development snapshots.

Result

Safe rollback and backup workflow established.

Pending / Remaining Tasks

A. Installer & Licensing

Pending

- Create professional installer package.
- Add custom EXE installer.
- Implement installation key/license system.
- Display unique installation key inside application.
- Add installer branding.

Time required to complete: 2days

B. Advanced Telemetry Dashboard

Pending

- Improve dashboard styling.
- Add modern telemetry cards.
- Add warning indicators.
- Add dynamic battery color alerts.
- Add failsafe indicators.
- Improve responsive layout.

C. 3D Vehicle Visualization

Pending

- Investigate 3D vehicle visualization.
- Add roll/pitch/yaw visualization.
- Evaluate WPF/HelixToolkit integration.
- Add smooth rotation/interpolation.

Time required: 2 working days

D. Video Feed Integration

Pending

- Embed live video feed into application.
- Add VLC/GStreamer integration.
- Add RTSP/UDP support.
- Add drone video panel.
- Add future HUD/video overlay support.

E. Swarm Planning Enhancements

Pending

- Improve swarm path visualization.
- Add collision-aware planning.
- Add formation presets.
- Add synchronized mission management.
- Add swarm simulation/testing.

F. UI/UX Improvements

Pending

- Final UI polish.
- Improve spacing/alignment.
- Add modern icons.
- Improve dark theme consistency.
- Add animations/transitions.

G. Stability & Testing

Pending

- Full regression testing.
- Startup stability improvements.
- Memory optimization for image rotation.

- Error handling improvements.
- Build validation.
- Deployment testing.

Current Technical Highlights

- Dynamic telemetry dashboard
- Swarm mission generation
- Multi-plan waypoint handling
- Real-time telemetry binding
- Vehicle orientation visualization
- Enhanced map interaction
- Backup workflow
- Modular tab-based UI redesign

Next plan

Short-Term

1. Finalize telemetry dashboard UI.
2. Stabilize splash screen and startup flow.
3. Complete installer/license system.
4. Improve swarm visualization.
5. Add professional video feed panel.

Medium-Term

1. Add 3D orientation visualization.
2. Add live drone stream support.
3. Add advanced swarm controls.
4. Improve mission synchronization.

Development Notes

- Existing Mission Planner telemetry architecture is reused via bindingSourceHud.
- Current implementation avoids duplication of telemetry parsing logic.
- Most modifications focus on presentation, usability, and operational workflow enhancements.
- Backup process established for safer development.